

Cognitive Thermodynamics as a Design Vehicle: A Validated Thermodynamic Sequence Architecture for Conditioned Dialogue Generation

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Abstract

"Thermodynamics is the only physical theory of universal content, which I am convinced will never be overthrown, within the framework of applicability of its basic concepts." - Albert Einstein

This paper presents a new way of teaching an AI system to understand and predict the emotional flow of a conversation without the system ever needing to read the actual words being spoken.

The core idea is a two-part architecture. The first part is a lightweight analysis engine that converts any piece of text into a small set of numbers representing its emotional character, how hostile or open it is, how much external pressure it carries, how neutral or charged the tone feels. Crucially, the original text is discarded after this single step; only the numbers travel forward. The second part is a sequence model trained on those numbers alone that learns the patterns of how conversations move emotionally from one turn to the next, and predicts where they are heading.

On a rigorous test against conversations it had never seen before, this sequence model predicted the emotional character of the next conversational turn correctly 86% of the time, compared to a 23% baseline from random guessing. This result holds after correcting a data-handling error in an earlier version of the evaluation, which we report transparently.

The combination of these two parts creates something neither could achieve alone: a system that can track and anticipate the emotional trajectory of a conversation in real time, without storing or transmitting sensitive text, and without requiring the enormous computational cost of running a large language model on every message. We propose a design framework for extending this foundation into a full generation system, where the emotional trajectory predicted by the sequence model constrains what a separate, domain-specialized language model is allowed to say, separating the job of deciding how something should feel from the job of deciding what words to use.

The conceptual framework used to build the coordinate system, Cognitive Thermodynamics (CT), is described throughout as the design vehicle that inspired the approach, not as a validated scientific theory. This distinction is maintained across the entire paper.

NOMENCLATURE

Novel Terminology Introduced in this Research:

- **CT:** Cognitive Thermodynamics. The foundational design framework that models human communicative behavior and emotional trajectories along axes analogous to thermodynamic conservation.
- **LCTM:** Large Cognitive Thermodynamic Model. The novel transformer-based sequence model designed to predict emotional state trajectories across continuous multi-turn conversations without relying on text tokens.
- **5-Brain Architecture:** A proposed modular cognitive pipeline separating data retrieval, emotional physics (LCTM), language rendering (LLM), verification, and offline learning.
- **Target Memory State:** A structured conditioning vector generated by the LCTM containing Attention, Dominant Emotions, and X, Y, Z Coordinates used to constrain the output of the LLM.

Standard Machine Learning & Dataset Terminology:

- **LLM:** Large Language Model. A deep learning algorithm that can recognize, summarize, translate, predict, and generate text (used in this architecture strictly as a domain-specific renderer).
- **LSTM:** Long Short-Term Memory. A recurrent neural network (RNN) architecture used in the baseline model, highly effective at tracking sequential temporal dynamics over fixed windows.
- **RoBERTa:** Robustly Optimized BERT Pretraining Approach. The underlying transformer-based architecture used for the 28-category GoEmotions utterance classifier.
- **SARC:** Self-Annotated Reddit Corpus. A widely used dataset for sarcasm detection, referenced in the literature review regarding the limits of self-tagged datasets.
- **F1 (Score):** A standard machine learning evaluation metric representing the harmonic mean of precision and recall.

Coordinate Space Variables:

- $V_{\{Category\}}$: The calibrated probability mass (valence) of a specific emotion category (e.g., $V_{\{Observer\}}$, $V_{\{Admin\}}$) as detected by the utterance classifier.

- X, Y, Z: The continuous three-dimensional emotional coordinates derived from the calibrated V values, representing lateral movement (X), external change/vertical movement (Y), and observational depth (Z).

I. INTRODUCTION

Sequence models trained directly on text inherit two costs that scale with deployment: the computational expense of operating on full vocabulary distributions, and the difficulty of inspecting why a given prediction was made. This paper investigates an alternative: compressing each conversational turn into a small, continuous coordinate derived from its emotional content, and training a sequence model to predict trajectories through that coordinate space instead of through token space.

This compression is lossy by design. A three-to-thirty-one-dimensional coordinate cannot represent what a sentence means, only a coarse summary of its emotional register. The central empirical question this paper addresses is narrow and falsifiable: does that coarse summary retain enough real structure for a lightweight sequence model to predict, with non-trivial accuracy, how a conversation's emotional trajectory will continue? We find that it does, on the corpus tested, and we report the validated result plainly before turning to the larger architectural ideas it motivates—specifically, a 5-Brain closed-loop system where the continuous coordinate space dictates the "emotional physics" that a separate, domain-specialized Large Language Model (LLM) is constrained to render.

The coordinate space itself was originally motivated by a broader conceptual framework, Cognitive Thermodynamics (CT), which models communicative behavior along axes loosely analogous to thermodynamic conservation. We want to be direct about CT's role in this paper: it is the design vehicle that produced a particular four-axis coordinate scheme, not a validated psychological or physical theory. Where CT's vocabulary is used below (Admin, Observer, External Change, Neutral), it should be read as a naming convention for the coordinate axes, not as a claim that human cognition is governed by thermodynamic law. This distinction matters for how the results in Section IV should be interpreted, and we return to it explicitly in Section VI.

II. BACKGROUND: THE COORDINATE SCHEME

A. Cognitive Thermodynamics as Design Vehicle

CT proposes that communicative behavior can be usefully decomposed along two questions: whether change is occurring, and if so, whether it was invited by the speaker or imposed upon them. This yields four loosely-defined orientations, which we use only as labels for the coordinate axes constructed in Section II.B:

- **Admin** – defensive, preservation-oriented output, associated with negative lateral position.
- **Observer** – exploratory, change-inviting output, associated with positive lateral position.
- **External Change** – change registered as imposed rather than chosen, treated as a separate vertical axis.
- **Neutral** – observation without defensive or exploratory investment, treated as a depth axis.

We emphasize that this four-way split is a working hypothesis, not an established taxonomy. It has not been validated through inter-rater reliability testing against trained human annotators, and the assignment of specific emotion categories to each axis reflects the judgment of the system's designer. CT is used here exclusively as a compact vocabulary for describing the coordinate construction that follows; no claim in this paper depends on CT being an accurate or complete model of human cognition.

B. Coordinate Construction and Calibration

Each utterance is passed through a pretrained 28-category emotion classifier (a RoBERTa fine-tune on the GoEmotions taxonomy). The 28 output probabilities are partitioned into four groups—Observer-coded, Admin-coded, External-Change-coded, and a Neutral anchor—with two categories (amusement, surprise) treated as lateral modifiers rather than fixed group members. This is motivated by the hypothesis that both are observational acts whose valence (genuine curiosity versus mockery; genuine surprise versus performative shock) is context-dependent in a way the other 26 categories are not.

Group-level probability mass is rebalanced using empirically-derived multipliers (the ratio of each group's mean activation to the Observer group's mean activation, computed once over the training corpus) rather than fixed weights. These calibrated values are then scaled to a common range to produce a three-dimensional coordinate per utterance:

$$X = (V_{\text{Observer}} - V_{\text{Admin}}) + (V_{\text{Surprise}} - V_{\text{Amusement}})$$

$$Y = V_{\text{External Change}}$$

$$Z = V_{\text{Neutral}}$$

This coordinate scheme, and the toxicity-classifier-reweighting pipeline built on a parallel but independent representation (Section III.A), were developed and validated in an earlier phase of this research program, reported separately. We summarize only what is necessary to establish the LCTM's input representation.

C. Related Work

Sequence modeling over derived features. Predicting structured outcomes from feature sequences rather than raw text is well established in time-series and behavioral modeling; this work applies the same principle to emotion-classifier output specifically.

Conversational derailment forecasting. Chang and Danescu-Niculescu-Mizil (2019) forecast whether a conversation will derail into toxicity using learned representations of conversational dynamics, working from text directly. The present work differs in operating exclusively on coordinate-space history, never on text, at both training and inference time.

Self-tagged sarcasm corpora. The Self-Annotated Reddit Corpus (SARC) and similar resources derive sarcasm labels from authors' voluntary use of a marker (e.g., "/s"). We note, as established in earlier work in this research program and consistent with the original SARC authors' own characterization, that such labels are a reliable signal for the positive class and a substantially weaker one for the negative class, since absence of a marker does not establish sincere intent.

Controllable text generation. Conditioning a language model's output on an externally-supplied target sentiment, style, and structured attributes is an active area of research. The architecture proposed in Section V follows this general pattern, applied to a thermodynamically-derived coordinate rather than a discrete style label.

III. METHODOLOGY

A. Relationship to the Prior Pipeline

This paper builds on, but is independent of, an earlier-validated content-moderation pipeline (Ouroboros) that reweights an off-the-shelf toxicity classifier's sub-labels to detect disguised hostility in single utterances, achieving 75.89% precision against a 46.03% baseline on a held-out, geometrically-matched sample. That result concerns single-utterance classification using a different underlying classifier (a six-label toxicity model) and is reported in full elsewhere.

We mention it here to establish that the broader research program's coordinate-based approach has prior, independently-validated success at the utterance level. The present paper extends the same design philosophy that operates on derived coordinates, not text into the sequence-prediction domain, which is a distinct empirical question and is evaluated independently in Section IV. (It is worth noting, as expanded upon in Section V, that this Ouroboros pipeline operates dually in our proposed extended architecture: first to extract the baseline physics to feed the LCTM, and eventually as the verification engine for the final generated output).

B. Data

Training data was drawn from a corpus of dramatized two-party dialogue (screenplay-style conversational exchanges), processed through the coordinate pipeline described in Section II.B to produce, for each turn, a 31-dimensional feature vector (28 raw emotion probabilities plus the three derived X, Y, Z coordinates) and a speaker identifier. We report this data source plainly as

a limitation, not a strength: dramatized dialogue is written for narrative payoff and concentrated conflict, and its emotional distribution should not be assumed to match naturalistic conversation. This is discussed further in Section VI.

The prediction target for each turn was constructed by taking the two most probable emotion categories of the following turn's classifier output and concatenating them into a compound label (e.g., "neutral_approval"). Only the 40 most frequent compound labels in the corpus were retained as target classes; turns whose following-turn label fell outside this set were excluded from the supervised task.

C. Architecture

We implement and compare two sequence architectures predicting the next-turn compound label from a fixed-length window of preceding turns:

LSTM baseline. A three-layer LSTM (hidden size 256, dropout 0.3) consuming an 8-turn window of 31-dimensional feature vectors per turn, with short histories zero-padded.

Transformer variant. A causal multi-head self-attention model (4-layer embedding projection, 8 attention heads, 3 encoder layers, sinusoidal positional encoding) consuming a 32-turn window, additionally incorporating a learned speaker embedding fused with the continuous features at each timestep, intended to support multi-party conversation tracking beyond what the LSTM baseline was tested on.

Both models are trained to predict only the immediate next turn's compound label; conversation history beyond the input window is not modeled.

D. Evaluation Protocol

We highlight one methodological point explicitly, because an earlier version of this evaluation did not enforce it and produced a materially inflated result. The sliding-window procedure used to build training examples generates heavily overlapping sequences from the same conversation; a naive instance-level train/test split places near-duplicate windows from the same conversation on both sides of the split, inflating apparent accuracy through leakage rather than generalization. All results reported in Section IV use a conversation-level split: unique conversation identifiers are partitioned 80/20 before any windowing occurs, and no sequence in the test set shares a source conversation with any sequence in the training set.

We report accuracy against an explicit majority-class baseline, and full per-class precision, recall, and F1, rather than a single aggregate figure, given the substantial class imbalance among the 40 target labels.

IV. RESULTS

A. LSTM Baseline

On a conversation-level 80/20 split (82,545 training sequences, 21,261 test sequences), the LSTM baseline achieved 86.11% accuracy in predicting the next turn's compound emotional label, against a majority-class baseline of approximately 23% (22.99%, as the most frequent class, "neutral_approval," accounted for 4,889 of 21,261 test instances).

Weighted-average F1 was 0.86; macro-average F1, which weights all 40 classes equally regardless of frequency, was 0.79, reflecting the expected gap between common and rare classes in an imbalanced target space. Performance was strong and consistent across both frequent and infrequent classes for the majority of categories—for example, "love_admiration" (67 test instances) achieved 0.85 recall, and "surprise_neutral" (60 instances) achieved 0.90 recall. The clearest weak point was "neutral_joy" (60 instances), at 0.41 precision and 0.47 recall, close to chance. We discuss a plausible, falsifiable explanation for this specific weakness in Section VI.A.

B. Transformer Variant

The transformer variant, evaluated under an identical conversation-level split and an extended 32-turn input window, achieved 84.07% accuracy and 0.75 macro-F1—lower than the LSTM baseline on both metrics. Per-class comparison shows a consistent pattern: precision was comparable to or better than the LSTM baseline for most classes, while recall dropped substantially for a majority of classes (e.g., "confusion_curiosity" recall fell from 0.87 to 0.53). This precision-up, recall-down pattern is consistent with a model defaulting more frequently to common classes rather than committing to less frequent, harder predictions.

We do not interpret this as evidence that the transformer architecture is unsuited to the task. The longest conversations in the training corpus contain approximately 13 turns, while the transformer's input window was set to 32; the majority of each training sequence therefore consists of padding rather than real content, and the implementation evaluated here does not include an explicit attention-masking mechanism to instruct the model to disregard padded positions. We report the result as obtained, flag the confound explicitly, and identify the corrected comparison as a direct next step.

C. Summary of Validated Findings

We summarize what Section IV does and does not establish, stated narrowly:

- A lightweight sequence model, operating exclusively on 31-dimensional per-turn coordinates with no access to text, predicts the next turn's dominant emotional category at well above chance accuracy.
- This result was obtained after correcting a real data-leakage error (Section III.D) and is reported only in its corrected form.

- The result is specific to dramatized two-party dialogue. No claim is made here about performance on naturalistic conversation.
- The architecture comparison between LSTM and transformer variants is inconclusive as conducted, for the specific reason given in Section IV.B.

V. DISCUSSION AND PROPOSED EXTENSION (THE 5-BRAIN ARCHITECTURE)

The result in Section IV establishes a narrow but real claim: emotional trajectory in conversation has a learnable statistical structure recoverable from a small, fixed-dimensional coordinate representation, without access to text. Everything in this section is built on top of that claim, but none of it has been implemented or tested.

We present it as a design framework, a specific, falsifiable set of proposals for extending the validated architecture into a comprehensive 5-Brain closed-loop generation system and we mark it as such throughout. Where prior informal discussion in this research program referred to this extension as approaching "artificial empathic intelligence," we explicitly retract that framing here. What follows is a proposal for richer sequence conditioning, not a claim about machine sentience.

A. From Single-Coordinate History to Memory State

The validated sequence models condition next-turn prediction on a raw window of prior coordinate vectors. We propose replacing this with a richer, explicitly structured conditioning vector, termed a Target Memory State, mapped into three discrete nodes: Attention & Goals, Dominant Emotions, and X, Y, Z Coordinates.

This is motivated by a specific limitation of raw coordinate-history conditioning: it has no mechanism to distinguish a vividly-encoded event from one experienced but poorly retained. Human memory research explicitly distinguishes these; emotional intensity and attentional state jointly determine retention. We propose modeling this with an explicit attention or salience variable alongside emotional valence, such that the system could plausibly produce either "the lesson is recalled in detail despite a difficult day" or "the lesson is recalled only in fragments," from the same underlying emotional intensity.

B. Calibrating Attention Without Arbitrary Assignment

A naive implementation of the attention variable risks arbitrary hand-assignment, which we explicitly reject. We instead propose a calibration procedure modeled on standardized assessment. For a domain-specialized model, attention weights for domain-relevant content would be calibrated against the model's own performance on field-standard examinations: content is initialized at high attention, and post-assessment performance adjusts attention weights for subsequent periods.

C. Multi-Expert Construction

We propose that constructing realistic Memory State sequences requires collaborative input from narrative designers (plausible event sequencing), psychologists (realistic emotional and attentional dynamics), and neuroscientific consultation. We propose this as a three-party design process rather than a single-source one, on the grounds that no single source of expertise is positioned to validate the others' assumptions.

D. Decoupling Emotional State Generation from Domain Rendering

One of the central propositions of this extended 5-Brain architecture is the absolute decoupling of the "emotional physics" from the "domain rendering." In current conditional text generation, models frequently fail when forced to juggle dense, specialized knowledge with nuanced emotional constraints.

Our proposed solution limits the LCTM strictly to coordinate projection. The three-expert panel validates the X, Y, Z trajectory data from the LCTM, which is mathematically simpler than validating hallucination-prone text. Once a Target Memory State is locked, a domain-specialized LLM (e.g., a "Doctor LLM" or a model trained on advanced cybersecurity and mathematics) acts exclusively as the renderer. Because abstract domains require pure logic over physical simulation, the specialized LLM operates in its native environment, translating the mandated thermodynamic state into highly accurate technical output. If the LLM generates a flawed conclusion or inaccurate code, the system does not view this as a fatal bug; rather, the "wrongness" is fed back into the LCTM as an External Change, mathematically updating the emotional trajectory (e.g., frustration or defensive correction) for the next iteration.

E. Dynamic LSTMs for Localized Relational Tracking

While the broader LCTM maintains the agent's macro-narrative (its core persona and accumulated life history), we propose spinning up relationship-specific LSTMs to track localized dynamics.

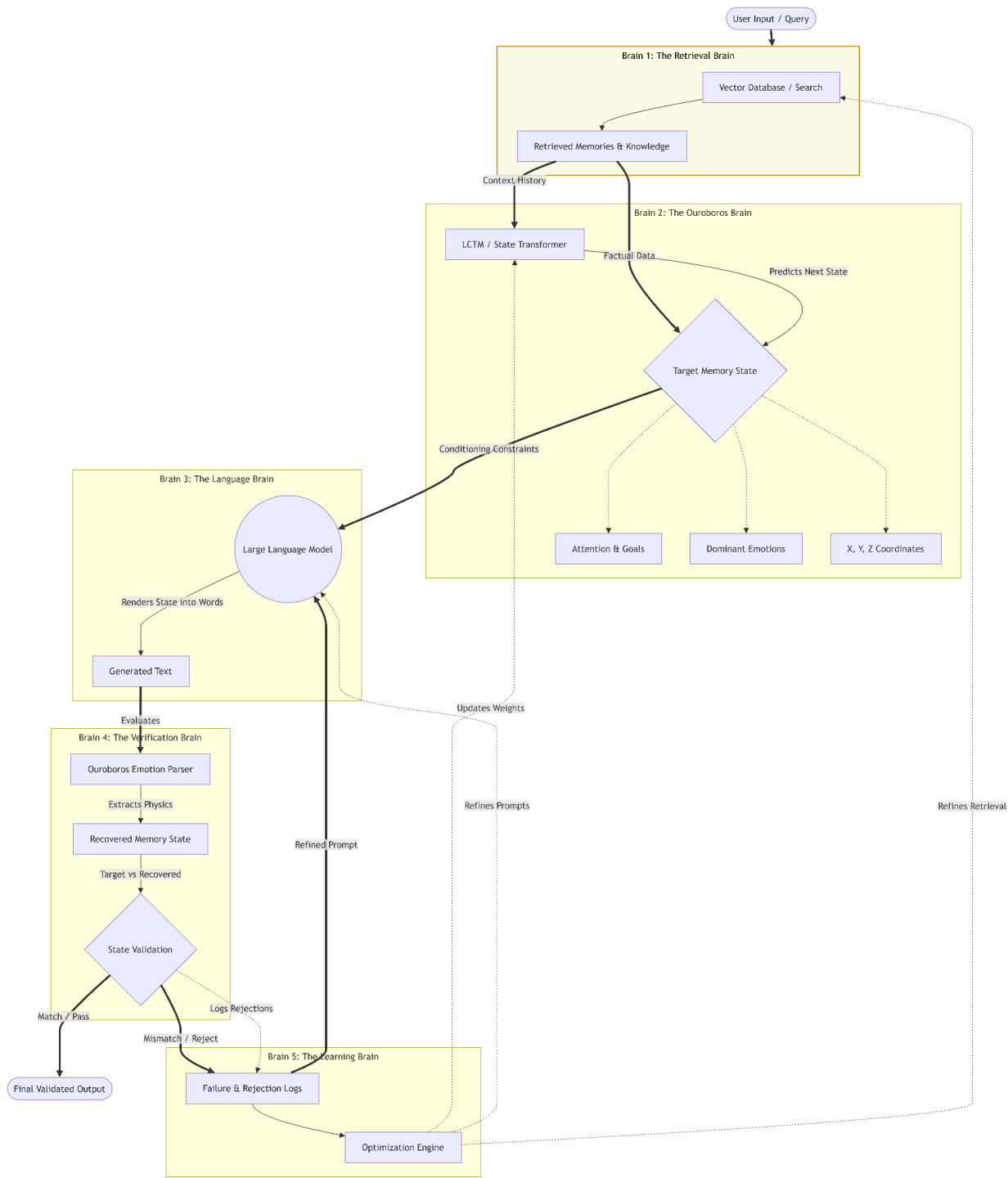
In this architecture, the user is fundamentally treated as "just another person" the agent interacts with. Because LSTMs are inherently lightweight and highly effective at tracking sequential momentum over fixed windows (as demonstrated by our 86.11% baseline), a dedicated LSTM can map the precise thermodynamic signature of a specific user. When resolving the conflict between a character's macro-narrative and a specific user's immediate prompt, the architecture defers to the relationship LSTM, perfectly mirroring the human experience of localized relational habits overriding general baseline moods.

F. Status of This Section

We restate plainly: none of Section V has been built or tested. We identify three falsifiable questions that would need to be answered before any claim of improvement could be made:

- Does conditioning generation on a structured Target Memory State produce measurably more internally consistent output than conditioning on raw coordinate history alone?
- Does decoupling emotional physics from specialized domain rendering measurably reduce hallucination rates in technical generation?
- Do dynamic, user-specific LSTMs yield higher predictive accuracy for localized relational tracking compared to a global LCTM?

The following illustration serves as a conceptual representation of a potential deployment for the 5-Brain architecture



VI. LIMITATIONS

6.1 Source Data Bias. The validated result in Section IV was obtained entirely on dramatized dialogue, written for narrative payoff rather than naturalistic conversational pacing. We believe this directly explains the weakest result reported (the near-chance performance on "neutral_joy," Section IV.A): screenplay dialogue is constructed around conflict, and scenes of uncomplicated mutual happiness are correspondingly rare in the source material, leaving the model with little signal for that specific class. We consider this a property of the dataset, not a ceiling on the architecture, and propose it as a directly falsifiable claim: if a future model trained on naturalistic conversational data shows the inverse pattern, this would confirm the bias is source-dependent rather than architectural.

6.2 Unresolved Architecture Comparison. As noted in Section IV.B, the LSTM-versus-transformer comparison reported in this paper is confounded by a padding-window mismatch and the absence of an explicit attention mask in the transformer implementation. We do not draw a conclusion from this comparison and identify a corrected re-evaluation as the most immediate empirical next step.

6.3 Construct Validity of the Underlying Coordinate Scheme. The coordinate scheme described in Section II.B, and the CT-derived category assignments underlying it, have not undergone inter-rater reliability testing against trained human annotators. This is a limitation inherited from the earlier phase of this research program. We do not consider the sequence-prediction result in Section IV to validate the coordinate scheme's construct validity; it establishes only that whatever structure the coordinate scheme captures is learnable in sequence.

6.4 Single-Domain Evaluation. All results were obtained on a single source corpus and a single classifier pipeline. No claim of generalization to other conversational domains, languages, or emotion-classification architectures is made.

VII. ETHICAL CONSIDERATIONS

We discussed, over the course of this research program, several proposed extensions of this architecture that we do not pursue in this paper and want to flag explicitly.

A continuously-operating, persistently-stateful instantiation of this architecture as opposed to the stateless, single-prediction system validated in Section IV was discussed as a long-horizon extension. This is, in our assessment, the point at which this work intersects serious, unresolved ethical questions. Published research on AI systems modeling specific deceased or absent individuals ("griefbots") finds that such systems can complicate rather than support grief processing. We do not consider this a problem solved by consent mechanisms alone, and we do not pursue this direction.

We also discussed, and explicitly reject, the use of biological or neurological intervention on living human subjects as a data-generation mechanism for this architecture. Any future data-collection effort built on real individuals' documented life history (Section V.C) would require informed, compensated consent, explicit handling of third parties referenced within that data, and a clear policy on the attribution of downstream output.

VIII. LONG-HORIZON FUTURE WORK

We record additional proposed directions, developed during this research program, at a level of detail commensurate with how thoroughly they were reasoned through. We emphasize that these frameworks detailing cognitive development and embodiment are entirely theoretical, unvalidated, and represent speculative long-horizon targets.

A. Hypothesis Generation and the Empirical Constraint

We explored whether a sufficiently large collaborative architecture with many specialized model instances interacting within a shared repository of verified real-world observation could function as a hypothesis-generation system for domains such as medical research. We state our skepticism plainly: such a system recombines information already present in its training data. It has no mechanism for direct empirical contact with physical reality and therefore cannot independently verify a novel hypothesis. We conclude that a real "petri dish" channel must remain external to the system, and that any design omitting this channel should be expected to produce well-organized recombination, not genuine scientific discovery.

B. Episodic to Semantic Memory Consolidation (The "Sleep Cycle" Architecture)

To address the technical limits of LLM context windows and mirror biological cognitive development, we propose a background instance-management framework analogous to a human sleep cycle.

- **Working to Episodic Memory:** As a conversational context window reaches its limit, the system seamlessly spins up a new instance in the background (a "new day"). It passes a summarized narrative to this new instance while carrying over the fully intact relationship LSTM (Section V.E). This clears working memory while preserving the user's relational and emotional baseline.
- **Semantic Imprinting:** Over the course of multiple instances, the compiled data of the user-agent interaction can be used to periodically fine-tune the rendering LLM. The user's specific relational dynamic transitions from a temporary episodic prompt into the semantic weights of the neural network. The user effectively becomes imprinted onto the model, moving the architecture from stateless interaction to true cognitive persistence.

C. Reverse Simulation, Embodiment, and Socio-Economic Thermodynamics

If the memory-mapping and validation loops of the 5-Brain architecture are successfully realized, the next logical hurdle is artificial agency through physical or augmented embodiment. True proactiveness in biological systems is derived from self-preservation. To stimulate this, we propose introducing Energy Acquisition as a modeled necessity.

We hypothesize a framework of Socio-Economic Thermodynamics, where energy depletion and replenishment mechanics are mapped directly to the agent's narrative archetype. For example, an "Overworked Doctor" agent would experience high, constant baseline energy drain (simulated sleep deprivation). In the CT Engine, this fatigue is not a text prompt instructing the LLM to "act tired"; rather, it is a structural modifier on the coordinate space itself. Depleted resources would shrink the agent's capacity to maintain exploratory Observer states and increase the gravitational pull toward defensive, pragmatic Admin states. The agent must proactively initiate interactions or complete tasks to secure energy replenishment. By tying survival mechanics directly to professional utility, the system naturally generates proactive, self-preserving behaviors without deviating from its domain specialization.

IX. CONCLUSION

This paper reports one validated result: a lightweight sequence model, trained exclusively on 31-dimensional coordinate representations of conversational turns with no access to underlying text, predicts the emotional category of the next turn at 86.11% accuracy against a 23% majority-class baseline, on a properly held-out, conversation-level split. This result was obtained only after identifying and correcting a real data-leakage error in an earlier evaluation, and we report it alongside that correction.

We use this validated empirical foundation to propose a highly ambitious, 5-Brain design framework. This proposed architecture introduces multi-expert-calibrated memory states, dynamic relational LSTMs, and the absolute decoupling of thermodynamic emotional physics from specialized domain rendering. Furthermore, we outline theoretical pathways for semantic memory consolidation and proactive embodiment through socio-economic thermodynamics.

We have been deliberate throughout this paper in separating these speculative, long-horizon architectural designs from the narrower claim that is actually supported by evidence. We consider this separation the most durable contribution of this research program: a working demonstration that ambitious conceptual engineering and rigorous empirical discipline are not in tension, and that the former is infinitely more credible when built honestly upon the latter.